QDP-1: A Multi-Stage Framework for Detecting Quantum Spin Coherence in Radical Pair Systems

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Abstract

We propose QDP-1, an eight-stage detection protocol for identifying quantum spin coherence effects in radical pair systems under ambient conditions. The framework makes three methodological contributions: (1) it unifies angular dependence testing, field-strength power-law discrimination, temperature-dependent coherence scaling, RF disruption testing, and negative control gating into a single scored detection framework for radical pair systems; (2) it adapts the principle of deliberate noise amplification from zero-noise extrapolation (ZNE) in quantum computing as a decoherence decay diagnostic–though this component requires experimental validation to determine whether the non-linear regime is practically accessible; and (3) it introduces specificity-weighted scoring with explicit control gating, adapted from particle physics discovery criteria (5σ thresholds) and clinical trial methodology. We validate QDP-1 in simulation (Stages 1–7) across four radical pair candidate systems and one known-negative control (a non-radical-pair fluorophore), demonstrating that the protocol differentiates between systems and that the control gate correctly caps verdicts when artifact contamination is present. Stage 5 (decoherence decay shape) scores 0/1 for all systems in simulation, consistent with the theoretical prediction that Lorentzian and linear decay converge in the small-perturbation regime. Stage 8 (RF disruption) is not simulated and requires experimental validation. No existing framework in the quantum biology literature unifies these elements into a single system-agnostic protocol. QDP-1 is validated in simulation only; experimental validation is needed. Simulation code is available at `https://github.com/HighpassStudio/qdp1`.

I. INTRODUCTION

The radical pair mechanism is the leading candidate explanation for biological magnetoreception and has been confirmed in cryptochrome systems through fluorescence-detected magnetic field effects [1] and Earth-strength magnetic sensitivity in tightly bound radical pairs [2]. Experimental techniques for detecting radical pair quantum effects have matured considerably: magneto-fluorescence fluctuation microspectroscopy enables investigation of quantum effects in biology at the single-molecule scale [3], and sensing magnetic-field effects at the scale of small molecular ensembles is now achievable [4].

Despite this progress, the field lacks a unified detection framework. Individual techniques–fluorescence-based magnetic field effect measurement, angular dependence characterization,

$B_{1/2}$ field-strength analysis, temperature-dependent coherence studies–are applied independently and evaluated by different criteria across different laboratories. This creates several problems:

1. **No standard for sufficiency.** How many positive indicators constitute evidence for quantum spin coherence? There is no consensus threshold analogous to the 5σ standard in particle physics.

2. **No control gating.** Negative controls are used, but there is no formal mechanism to prevent false positives from propagating when controls show contamination.

3. **No cross-disciplinary diagnostic transfer.** Zero-noise extrapolation (ZNE), a technique developed for quantum computing error mitigation [5, 6], provides a natural framework for probing quantum coherence effects through deliberate noise scaling. This approach has begun migrating from quantum computing into quantum sensing [7] but has not been applied to quantum biology or radical pair systems.

4. **No system-agnostic protocol.** Each new radical pair candidate is evaluated with ad hoc methods. A reusable detection kit would accelerate the field.

QDP-1 addresses these gaps by combining eight partially independent tests into a scored verdict with explicit control gating. The protocol is designed to be applied identically to any radical pair system, with each application constituting an independent experimental contribution. This work is intended as a starting point for community evaluation and refinement.

A. Scope

QDP-1 detects signatures *consistent with* quantum spin coherence in radical pair systems. It does not test for entanglement specifically, nor does it address other quantum biological phenomena (coherent energy transfer, quantum tunneling) without additional stages. The framework provides evidence grades, not binary proof.

II. RELATED WORK

A. Radical Pair Detection Methods

The radical pair mechanism and its role in biological magnetoreception have been extensively reviewed [8, 9, 17]. Key experimental milestones include the detection of magnetic field effects on cryptochrome fluorescence [1], characterization of angular dependence in singlet yield [10], measurement of $B_{1/2}$ parameters for field-strength characterization [11], and investigation of temperature effects on spin coherence [12]. These techniques are individually well-established and are not claimed as novel contributions of this work.

B. Zero-Noise Extrapolation in Quantum Computing

ZNE was introduced as an error-mitigation technique for near-term quantum computers [5, 6]. The core idea is to deliberately amplify noise, measure the observable at multiple noise levels, and extrapolate to the zero-noise limit. ZNE is now standard practice in quantum computing and has begun migrating into quantum sensing–notably, a 2024 study applied ZNE-style extrapolation to mitigate errors in DC magnetometry [7]. However, the use of ZNE as a *diagnostic* (to distinguish quantum from classical behavior based on decay curve shape) rather than as an error-mitigation tool, and its application to biological or chemical radical pair systems, does not appear in the mainstream literature.

C. The Quantum Biology Detection Gap

The question "How quantum is radical pair magnetoreception?" has been addressed theoretically [13], establishing that radical pair magnetoreception satisfies formal criteria for quantum behavior from a quantum information perspective. However, this assessment is conceptual, not operational–it does not provide a staged detection protocol for experimental use.

The most relevant methodological advances are the 2024 magneto-fluorescence fluctuation microspectroscopy technique [3] and the study on sensing magnetic-field effects at small ensemble scales [4]. Both move toward reusable measurement methodology for quantum biology, but neither constitutes a multi-stage scored framework with control gating and

cross-disciplinary diagnostic integration.

A 2024 study on simulating spin biology using a digital quantum computer [14] represents the most direct bridge between quantum computing and quantum biology in the existing literature. However, the connection there is different: the paper uses extrapolation ideas akin to ZNE for simulation accuracy, not as an experimental diagnostic for biological systems.

D. Novelty Claim

QDP-1 unifies ZNE-style noise scaling (adapted from quantum computing), Fourier angular power analysis (adapted from signal processing), power-law field discrimination, temperature coherence scaling, and negative control gating (adapted from particle physics and clinical trials) into a single scored detection framework for quantum biology. Based on a search across 2005–2025 literature in major journals and arXiv, no mainstream precedent was found for this full combination. The likely novelty is the assembly, not the individual parts.

III. METHODS: THE QDP-1 PROTOCOL

QDP-1 consists of eight stages, each contributing a weighted score to a final verdict. Stage 3 (negative controls) serves as a gate: if controls fail, the final verdict is capped regardless of total score. The protocol requires a fluorescence-based readout of radical pair reaction yield as a function of applied magnetic field angle and strength.

A. Stage 1: Baseline Noise Characterization (1 point)

Purpose. Determine whether the measurement setup is shot-noise-limited before signal measurement begins.

Method. Measure N fluorescence pulses with no applied magnetic field. Compute the excess noise ratio:

$$\mathrm{ENR} = \frac{\sigma_{\mathrm{meas}}}{\sqrt{\mu}} \tag{1}$$

where μ is the mean detected counts and σ_{meas} is the measured standard deviation.

Decision. ENR < 1.5 passes (shot-noise-limited). ENR ≥ 1.5 flags technical noise dominance.

Scoring. 1 point if passed.

B. Stage 2: Differential Signal Detection (2 points)

Purpose. Detect angular dependence of radical pair yield using normalized differential measurement that cancels common-mode drift.

Method. Block-randomized alternation between field angle $\theta_A = 0$ and $\theta_B = \pi/2$. For each block of $2B$ pulses:

$$C_i = \frac{S_A - S_B}{S_A + S_B} \tag{2}$$

where S_A, S_B are summed photon counts at each angle.

Statistics. Mean contrast $\bar{C} = \langle C_i \rangle$, standard error SE $= \mathrm{std}(C_i)/\sqrt{N_{\mathrm{blocks}}}$, Z-score $Z = |\bar{C}|/\mathrm{SE}$.

Decision. $Z > 5$ for detection (5σ, matching the particle physics discovery threshold). $Z > 3$ for evidence.

Scoring. 2 points if $Z > 5$; 1 point if $Z > 3$.

C. Stage 3: Negative Controls–GATE (2 points)

Purpose. Eliminate artifacts. This stage is a gate: if controls fail, the final verdict is capped at "INCONCLUSIVE" regardless of total score.

Controls.

- C1: Same angle ($\theta_A = \theta_B = 0$). Tests for detector asymmetry.
- C2: No applied field / near-zero field. Tests for field-independent artifacts.
- C3: Killed radical pair (radical scavenger or denatured protein). Tests whether the signal requires the spin-correlated radical pair.

Decision. Each control must show $Z < 2.0$ (pass). $2.0 \leq Z \leq 3.0$ is indeterminate: the control neither passes nor fails, but the stage scores 1 point instead of 2 and the run should be flagged for replication. $Z > 3.0$ is a hard fail: artifact contamination is present and the gate activates.

Scoring. 2 points if all controls pass ($Z < 2.0$). 1 point if any control is indeterminate ($2.0 \leq Z \leq 3.0$) but none fail. 0 points if any control fails ($Z > 3.0$). Gate: failure caps verdict.

D. Stage 4: Angular Shape via Fourier Decomposition (2 points)

Purpose. Confirm that the angular dependence follows the $\cos^2\theta$ pattern predicted by radical pair theory.

Method. Measure yield at $N \geq 24$ equally-spaced angles from 0 to π. Compute the Fourier power in the 2θ component:

$$a_2 = \frac{2}{N}\sum_i y_i \cos(2\theta_i) \tag{3}$$

$$b_2 = \frac{2}{N}\sum_i y_i \sin(2\theta_i) \tag{4}$$

$$P_2 = a_2^2 + b_2^2 \tag{5}$$

Since $\cos^2\theta = 0.5 + 0.5\cos(2\theta)$, the radical pair signal concentrates power in the 2θ Fourier component.

Statistics. Define the total power as the sum over all non-DC Fourier components: $P_{\text{total}} = \sum_{k=1}^{N/2}(a_k^2 + b_k^2)$. The power ratio is $R_2 = P_2/P_{\text{total}}$. The F-statistic tests whether the 2θ component explains significantly more variance than expected under a flat (noise-only) null model: $F = P_2/(P_{\text{total}} - P_2) \times (N/2 - 1)$, with 2 and $N - 3$ degrees of freedom.

Decision. Power ratio $R_2 > 0.5$ AND $F > 10$ (corresponding to $p < 0.001$ for $N = 24$) for strong detection.

Scoring. 2 points if both criteria met. 1 point if either met alone.

E. Stage 5: Decoherence Decay Shape–ZNE-Inspired Diagnostic (1 point)

Purpose. Characterize the functional form of signal decay under controlled decoherence perturbation, as supporting evidence for or against coherent spin dynamics.

Method. Deliberately increase decoherence by controlled means (paramagnetic ions, temperature increase, or chemical modification). Measure differential contrast at decoher-

ence levels $d = 0.0, 0.1, \ldots, 0.9$. Fit three models:

$$\text{Linear:} \quad C(d) = a - b \cdot d \tag{6}$$

$$\text{Lorentzian:} \quad C(d) = \frac{a}{1 + k \cdot d} \tag{7}$$

$$\text{Stretched exponential:} \quad C(d) = a \cdot \exp\left[-(d/\tau)^{\beta}\right] \tag{8}$$

Rationale. This is inspired by ZNE from quantum computing [5, 6], where noise is deliberately amplified and the observable extrapolated to zero noise. Here, we adapt the principle as a *diagnostic*: the functional form of the decay curve under increasing decoherence differs between coherent spin dynamics and classical artifacts. However, we emphasize an important limitation: in quantum computing, noise can be amplified precisely via gate stretching. In biological systems, "increasing decoherence" via paramagnetic ions or temperature changes may simultaneously alter reaction kinetics, protein conformation, and radical pair lifetime. The controlled-parameter assumption is therefore approximate, and results from this stage must be interpreted cautiously.

Important caveats. (1) Non-linear decay alone is not proof of quantum coherence. Some classical processes (photobleaching, thermal relaxation) also produce non-linear decay. (2) The decoherence perturbation may not cleanly separate spin dephasing from other chemical effects. This stage provides supporting evidence within the multi-stage framework, not standalone proof. The scoring weight (1 point) reflects these limitations.

Decision. Use AIC to select among the three models. If the Lorentzian or stretched exponential is selected over linear ($\Delta\text{AIC} > 4$), the decay is classified as "non-linear (quantum-consistent)."

Scoring. 1 point if non-linear decay shape is selected by AIC.

1. Theoretical Basis

Under Markovian dephasing modeled via Lindblad operators, the off-diagonal elements of the spin density matrix responsible for singlet-triplet coherence decay as $\exp(-\gamma t)$, where γ is the dephasing rate. When integrated against the pair survival probability $\exp(-k_{\text{eff}} \cdot t)$, the time-averaged coherent modulation amplitude takes a Lorentzian form:

$$A(\gamma) \sim \frac{k_{\text{eff}}}{k_{\text{eff}} + \gamma} \tag{9}$$

Since the controlled decoherence parameter d scales the physical dephasing rate linearly–$\gamma(d) = \gamma_0 + c \cdot d$–the contrast follows:

$$C(d) = \frac{C_0}{1 + \alpha d} \quad \text{where } \alpha = \frac{c}{k_{\text{eff}} + \gamma_0},\ C_0 = \frac{k_{\text{eff}}}{k_{\text{eff}} + \gamma_0} \tag{10}$$

This Lorentzian decay is the correct functional form for a homogeneous radical pair system under controlled dephasing. For small perturbations ($\alpha d \ll 1$), the Lorentzian approximates $C_0(1 - \alpha d)$, which is linear–meaning the quantum and classical signatures converge in the weak-perturbation limit and are distinguishable only when αd is of order unity or larger. This places a practical requirement on Stage 5: the decoherence perturbation range must be large enough to enter the non-linear regime. For classical artifacts that scale proportionally with the perturbation (e.g., field-proportional detector coupling), the decay remains linear across the full range.

2. *Extension: Stretched Exponential for Heterogeneous Systems*

Real biological radical pairs embedded in proteins experience heterogeneous environments that produce a *distribution* of dephasing rates. When the rate distribution $\rho(u)$ is broad, the ensemble-averaged contrast becomes:

$$\frac{C(d)}{C_0} = \int_0^\infty \rho(u) \frac{1}{1 + u \cdot d} \rho(u)\, du \tag{11}$$

For certain rate distributions, this integral produces decay forms well-approximated by the Kohlrausch-Williams-Watts (KWW) stretched exponential:

$$C(d) \approx C_0 \cdot \exp\left[-\left(\frac{d}{\tau}\right)^\beta\right] \quad \text{where } 0 < \beta \leq 1 \tag{12}$$

The stretching exponent β encodes environmental heterogeneity: $\beta = 1$ corresponds to a narrow rate distribution (quasi-homogeneous); $0.5 < \beta < 1$ indicates heterogeneous decoherence environments typical of protein-embedded radical pairs. Note that stretched exponentials also arise in classical disordered systems (polymer relaxation, dielectric response), so detection of a stretched exponential is necessary but not sufficient for a quantum interpretation. The value of this diagnostic lies in its combination with the other stages, not in isolation.

F. Stage 6: Field-Strength Power Law (1 point)

Purpose. Distinguish quantum radical pair response from classical magnetic effects.

Method. Measure differential contrast at field strengths spanning ≥ 2 orders of magnitude. Fit power law to the weak-field regime:

$$\log(C) = \alpha \cdot \log(B) + \beta \tag{13}$$

Rationale. Quantum radical pair theory predicts $\alpha \approx 1$ at weak fields [11]. Classical effects scale as $\alpha \approx 2$.

Decision. $\alpha < 1.5$ indicates quantum-consistent scaling.

Scoring. 1 point if quantum-consistent scaling detected.

G. Stage 7: Temperature Coherence Scaling (1 point)

Purpose. Quantum spin coherence degrades with increasing temperature. Classical magnetic effects typically show different temperature dependence.

Method. Measure differential contrast at temperature offsets from baseline (-10 K to $+30$ K). Compute Spearman rank correlation between temperature and contrast magnitude.

Decision. Spearman $\rho < -0.5$ indicates quantum-consistent thermal scaling.

Scoring. 1 point if significant negative correlation detected.

H. Stage 8: RF Magnetic Field Disruption (2 points)

Purpose. Confirm radical pair involvement via resonant disruption of spin dynamics. Application of a weak radiofrequency (RF) magnetic field at the Larmor frequency should disrupt singlet-triplet interconversion in radical pairs, reducing the observed magnetic field effect [8]. This is widely regarded as one of the strongest diagnostics for radical pair involvement because no classical magnetic effect is disrupted by a weak RF field at a specific resonance frequency.

Method. Measure differential contrast (Stage 2 protocol) under three conditions: (1) static field only (baseline), (2) static field + RF at the Larmor frequency $\nu_L = \gamma_e B/2\pi$, and (3) static field + RF at a non-resonant frequency (control). Compute the RF suppression

ratio:

$$S_{\mathrm{RF}} = 1 - \frac{C_{\mathrm{RF}}}{C_{\mathrm{baseline}}} \tag{14}$$

Decision. $S_{\mathrm{RF}} > 0.2$ at the Larmor frequency AND $S_{\mathrm{RF}} < 0.05$ at the non-resonant frequency for strong detection. The frequency specificity is the key discriminator: classical effects are suppressed equally at all RF frequencies, while radical pair effects show resonant suppression.

Scoring. 2 points if resonant suppression detected with frequency specificity. 1 point if suppression detected but frequency specificity is marginal.

IV. SCORING AND VERDICT

A. Score Allocation

TABLE I. QDP-1 scoring allocation.

Stage	Test	Max points
1	Baseline noise quality	1
2	Differential signal detection	2
3	Negative controls (GATE)	2
4	Fourier angular shape	2
5	Decoherence decay shape	1
6	Field-strength power law	1
7	Temperature coherence scaling	1
8	RF magnetic field disruption	2
	Total	**12**

B. Verdict Thresholds

$\geq 10/12$: Consistent with quantum spin coherence. $\geq 8/12$: Strong evidence–recommend independent replication. $\geq 5/12$: Suggestive–additional data needed. $< 5/12$: No detection.

C. Scoring Weight Rationale

Stages are weighted by diagnostic specificity–the degree to which a positive result is uniquely attributable to radical pair quantum coherence rather than classical alternatives. Stages 2, 3, 4, and 8 receive 2 points each because they test properties with high specificity: angular dependence with 5σ significance (Stage 2), artifact elimination via controls (Stage 3), the characteristic $\cos^2\theta$ Fourier signature (Stage 4), and frequency-specific RF disruption (Stage 8). Stages 1, 5, 6, and 7 receive 1 point each because they test properties that, while consistent with quantum coherence, can also arise from classical mechanisms (noise characterization, non-linear decay, power-law scaling, temperature sensitivity). This weighting ensures that a system cannot reach a positive verdict on low-specificity stages alone.

D. Stage Independence

Stages 1–4 and 8 provide substantially independent lines of evidence. Stages 5–7 are partially correlated: temperature (Stage 7) affects decoherence rates (Stage 5), and decoherence affects field-scaling behavior (Stage 6). The scoring weights reflect this: the five high-independence stages carry 9 of 12 points, while the three correlated stages carry 3 points combined.

E. Control Gate

Stage 3 must pass for any positive verdict. If controls fail, the verdict is capped at "INCONCLUSIVE–ARTIFACT RISK" regardless of total score. This mechanism is adapted from blind-analysis protocols in particle physics and clinical trial gating.

V. SIMULATION VALIDATION

A. Systems Tested

QDP-1 was validated in simulation across four radical pair candidate systems and one known-negative control:

1. **Avian Cryptochrome Cry4a**–field ratio 0.05, 1000 photons/pulse.

2. **Flavin-Tryptophan Model**–field ratio 0.05, 2000 photons/pulse.

3. **Plant Cryptochrome Cry1**–field ratio 0.05, 800 photons/pulse.

4. **Enzyme Radical Pair**–field ratio 0.03, 500 photons/pulse.

5. **Rhodamine B (Negative Control)**–field ratio 0.0, 1500 photons/pulse.

B. Noise Model

Simulations include five realistic noise sources: Poisson shot noise, LED intensity drift ($\sigma = 0.5\%$), dark counts (20 counts/pulse), photobleaching (exponential decay, rate 0.1), and Gaussian detector read noise ($\sigma = 5$ counts).

C. Results

TABLE II. QDP-1 simulation results (Stages 1–7 only; representative run). Stage 8 (RF disruption) is not included in the simulation and would require experimental validation.

System	Score (of 10)	Verdict
Avian Cryptochrome Cry4a	7/10	Strong evidence
Flavin-Tryptophan Model	5/10	Suggestive
Plant Cryptochrome Cry1	7/10	Strong evidence
Enzyme Radical Pair	9/10	Consistent
Rhodamine B (Control)	3–4/10	No detection

Key observations: (1) Stage 2 cleanly separates RP from non-RP systems ($Z = 17.8$–95.0 for candidates vs. $Z = 0.4$ for control). (2) The negative control scores at the bottom (3–4/10). (3) Stage 5 (decoherence decay shape) scores 0/1 for all systems–the simulation's linear decoherence proxy produces linear decay, which is consistent with the corrected theoretical prediction that Lorentzian and linear forms converge in the small-perturbation regime (see Sec. III E 1). Distinguishing quantum from classical decay requires perturbations large

enough to enter the non-linear regime, which the simulation does not model. (4) Scores vary ±1–2 points between runs due to stochastic noise. (5) Stage 8 (RF disruption) is not simulated; it requires physical spin dynamics and RF field modeling beyond the scope of this simulation.

D. Simulation Limitations

The simulation uses a simplified decoherence model: linear field-ratio reduction as proxy for decoherence, uniform temperature coefficient, and Lorentzian-derived power-law scaling. The simulation validates *protocol logic*, not *physical predictions*.

VI. DISCUSSION

A. The Cross-Disciplinary Bridge

QDP-1 sits at the intersection of quantum computing (ZNE), spin chemistry (radical pair detection), and signal processing (Fourier analysis), importing methodology from particle physics and clinical trials. The contribution is the assembly, not new measurement physics.

B. The Decoherence Diagnostic: Status and Limitations

Stage 5 adapts the principle of deliberate noise amplification from ZNE in quantum computing, but the transfer is not straightforward. In quantum circuits, noise can be amplified precisely via gate stretching with a known scaling factor. In biological radical pair systems, "increasing decoherence" via paramagnetic ions or temperature changes may simultaneously alter reaction kinetics, protein conformation, and radical pair lifetime. The corrected theoretical analysis (Sec. III E 1) shows that the coherent contribution to the signal decays as a Lorentzian function of the decoherence parameter, which converges to linear decay in the small-perturbation limit–making it indistinguishable from classical artifacts unless the perturbation is large enough to enter the non-linear regime. Stage 5 scores 0/1 across all systems in simulation, which is consistent with this analysis: the simulation uses a linear decoherence proxy that does not reach the non-linear regime. Experimental validation with physical decoherence mechanisms (paramagnetic quenching at concentrations spanning at

least one order of magnitude) is required to determine whether the non-linear regime is practically accessible.

C. False Positive Analysis

A key question for any detection framework is: what is the probability that a purely classical system achieves a positive verdict? In the current simulation, the negative control (Rhodamine B) scores 3–4/10 on Stages 1–7, with ±1–2 point run-to-run variability. Under the revised 12-point scale, a classical system that passes Stage 1 (noise quality, likely) and accumulates points from Stages 5–7 (which can score on classical signals) could reach 3–4/12 before Stage 8. Stage 8 (RF disruption) provides a strong barrier: no classical magnetic effect shows frequency-specific RF suppression, so a classical system should score 0/2 on Stage 8. The combined effect of the control gate (Stage 3) and the RF disruption test (Stage 8) means that a classical system reaching the "suggestive" threshold of 5/12 would require both: passing controls AND scoring on multiple low-specificity stages simultaneously. A formal false positive rate calculation requires Monte Carlo simulation under realistic classical null models, which is beyond the scope of this work but is a priority for future validation.

D. Practical Application

Equipment requirements for optical/magnetic components are modest ($1,500–$4,000). However, biological sample preparation (protein expression, anaerobic handling, EM shielding) adds substantial infrastructure requirements. Priority targets: (1) Cry4a, (2) synthetic FAD-Trp models, (3) plant cryptochromes, (4) enzyme radical pairs.

E. Broader Implications

If validated experimentally, QDP-1 enables: quantum biology surveys across cryptochrome families, pharmacological screening for radical pair involvement [15], and investigation of radical pair mechanisms in oxidative stress and reactive oxygen species production [16].

VII. LIMITATIONS

1. **Simulation only.** Experimental validation needed.
2. **Simplified decoherence.** Stage 5 uses a proxy model.
3. **Uncalibrated scoring weights.** Need empirical validation.
4. **Partial stage correlation.** Stages 5–7 are correlated.
5. **Scope limitation.** Radical pair systems only.
6. **Non-linear decay caveat.** Not sufficient alone; classical processes can also produce non-linear decay.
7. **Missing standard tests.** Isotope substitution and EPR not included in the core protocol (proposed as extensions).
8. **Stage 4 false positives.** Fourier test can score on noise-only data.
9. **Run-to-run variability.** ±1–2 points between runs.

VIII. PROPOSED EXTENSIONS

Stage 9: Magnetic Isotope Substitution. Replacing magnetic nuclei (^{1}H $\rightarrow$ ^{2}H, ^{12}C $\rightarrow$ ^{13}C) alters hyperfine coupling predictably. Like RF disruption (Stage 8), this test has high specificity for radical pair involvement because no classical magnetic effect depends on nuclear spin.

Stage 10: Direct Radical Characterization via EPR/Transient Absorption. Time-resolved EPR confirms radical identity and provides direct observation of the spin-correlated radical pair.

IX. CONCLUSION

QDP-1 provides a structured, reproducible framework for evaluating quantum spin coherence signatures in radical pair systems. Its primary contribution is formalizing multi-stage evidence accumulation with control gating and RF disruption testing–a standardization that

does not currently exist in the field. The decoherence decay diagnostic (Stage 5) represents a speculative but potentially valuable cross-disciplinary transfer from quantum computing, requiring experimental validation to determine whether the non-linear Lorentzian regime is practically accessible in biological systems.

QDP-1 is a methodological proposal–a structured starting point for standardization–not a battle-tested detection kit. Its value will be determined by experimental validation, integration with established techniques, and collaborative stress-testing by the spin chemistry and quantum biology communities.

We invite collaboration from experimental groups working with radical pair systems.

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Simulation code and full paper are available at `https://github.com/HighpassStudio/qdp1`. This work benefited from AI-assisted analysis and literature review using Claude (Anthropic).

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